



# Measuring Militarization and the Link with Interstate Conflict

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Paper & Citations



## Overview

Prior research has focused on the relationship between changes in military expenditures/personnel and conflict (Colaresi & Thompson, 2005; Rider et al. 2011; Gibler et al. 2015). Militarization can both be a proxy for risk of conflict as well as having a causal effect on the probability of conflict. By separately measuring militarization stocks and flows, this paper attempts to isolate the causal effect of military strength. The paper finds that while militarization flows are positively associated with conflict, stocks are negatively associated with conflict, providing support for the deterrence hypothesis.

## Hypotheses

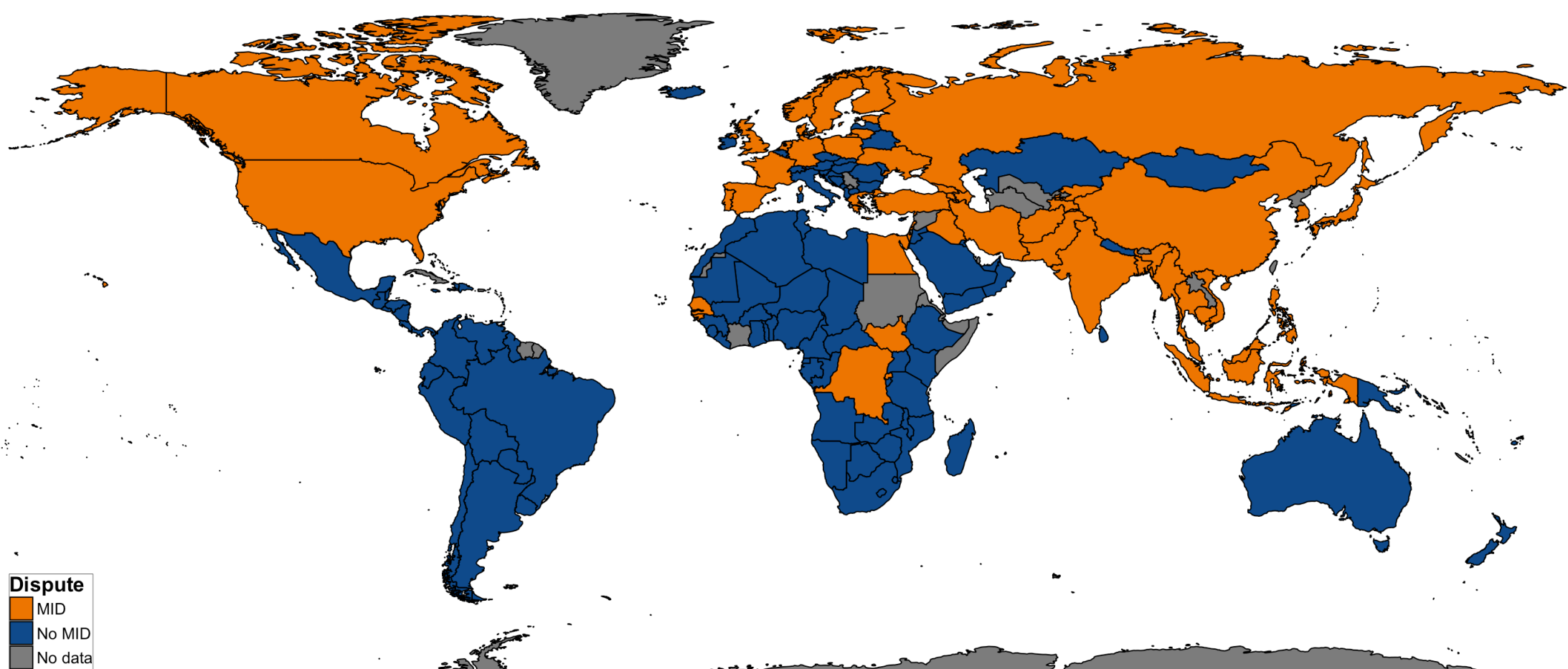
- H1:** Larger stocks of military capacity decrease the probability of militarized interstate disputes (MIDs) due to deterrence.
- H2:** Larger flows of militarization proxy for background conflict risk, and so are associated with greater probability of MIDs.

## Methods

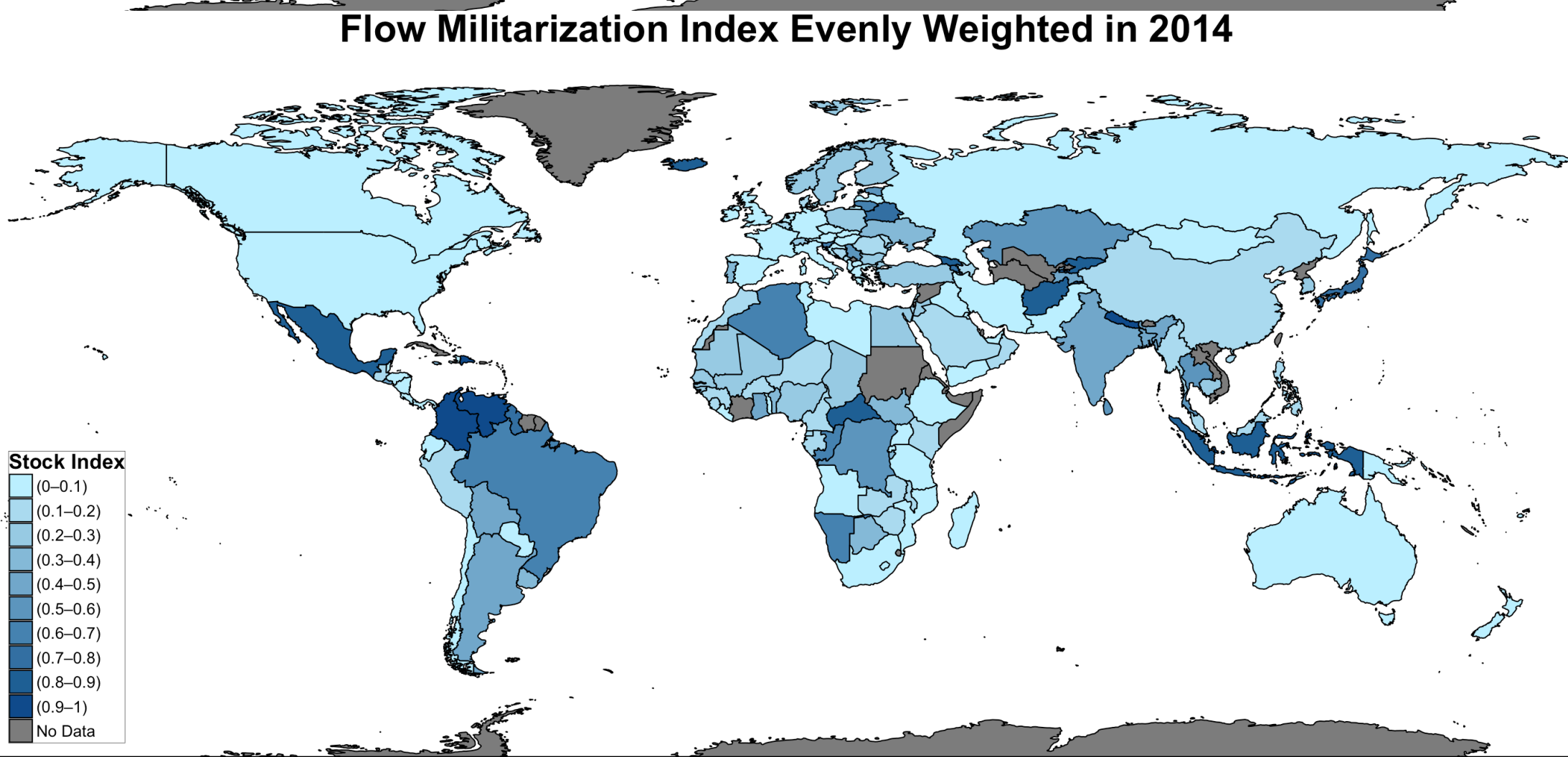
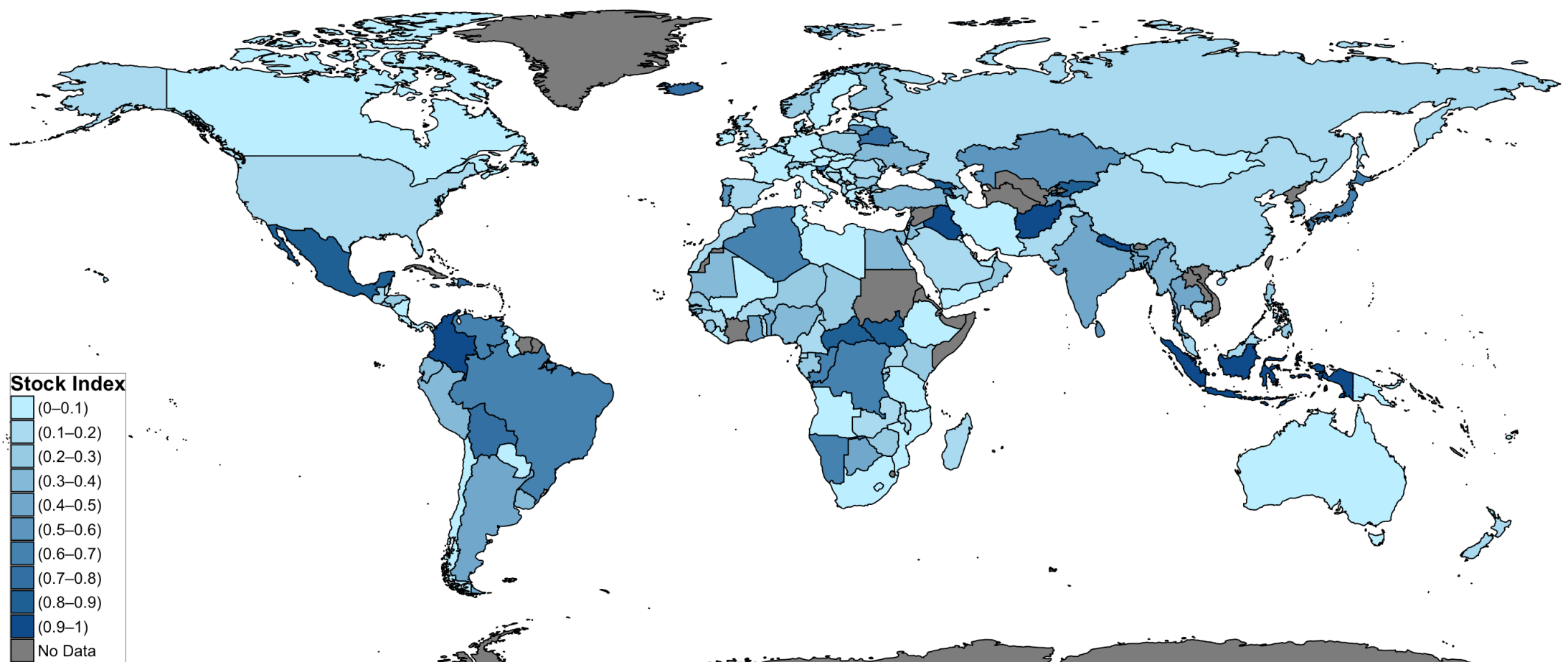
- Data is from 1960-2014 for 152 countries.
- Created stock and flow measures
  - Stock = Expenditure (2 years prior) + Military Personnel (2 years prior) + Weaponry (current)
  - Flow = Expenditure (current) + Military Personnel (current)
- Multivariate logistic regressions with imputation

## Data

**Militarized Interstate Dispute:** binary variable for if one state threatens, displays, or uses force against another state



**Militarization Index:** a combination of military expenditure, personnel and heavy weaponry  
Stock Militarization Index Evenly Weighted in 2014



## Results

|                        | Dispute            |                    | Originator         |                    |
|------------------------|--------------------|--------------------|--------------------|--------------------|
|                        | Model 1            | Model 2            | Model 3            | Model 4            |
| Stock Index            | -1.14**<br>(0.33)  | -1.32**<br>(0.35)  | -0.99**<br>(0.28)  | -1.14**<br>(0.30)  |
| Flow Index             | 1.44***<br>(0.30)  | 1.62***<br>(0.31)  | 1.31***<br>(0.25)  | 1.43***<br>(0.26)  |
| Democracy              |                    | 0.08<br>(0.05)     |                    | 0.07<br>(0.05)     |
| Economic Globalization |                    | 0.01**<br>(0.00)   |                    | -0.00<br>(0.00)    |
| Gender Equality        |                    | -1.31***<br>(0.19) |                    | -1.21***<br>(0.19) |
| Alliances              |                    | 0.07***<br>(0.01)  |                    | 0.06***<br>(0.01)  |
| Constant               | -0.89***<br>(0.06) | -0.85***<br>(0.12) | -1.08***<br>(0.07) | -0.67***<br>(0.12) |

\*\*\*  $p < 0.001$ ; \*\*  $p < 0.01$ ; \*  $p < 0.05$

## Key Findings

- Longer-term military investment decreases the risk of disputes and the initiation of disputes.
- Because short-term investment is positively associated with conflict, it likely is a proxy for the dangerousness of the current environment.

## Future Directions

- Use dyadic conflict data to analyze the impact of disparities in militarization between countries.
- Does the deterrence effect extend to when it is the allies that have higher militarization?